

CLINTONVILLE, WI, USA

WMO: 726502

Lat: 44.614N

Lon: 88.731W

Elev: 251

StdP: 98.35

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54832

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.1	-27.4	0.3	-21.0	-24.9	0.4	-19.0	9.3	-7.7	8.4	-8.2	2.1	290	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	32.0	23.2	30.0	21.7	28.0	20.7	24.5	29.1	23.5	27.9	22.4	26.7	4.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.2	26.9	22.3	17.5	26.2	21.2	16.3	25.1	75.6	28.9	71.0	27.7	67.1	26.5	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.7	6.9	DB	-26.2	34.6	2.3	2.1	-27.9	36.1	-29.3	37.3	-30.6	38.5	-32.3	40.1	(4)
(5)			WB	-26.4	26.2	2.3	1.8	-28.1	27.5	-29.4	28.5	-30.7	29.5	-32.4	30.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-7.5	-6.0	0.1	7.1	13.6	18.9	21.4	20.2	16.3	9.0	2.2	-4.7	(6)	
	DBStd	11.33	6.39	6.16	6.26	5.01	4.73	3.75	3.18	3.10	4.37	4.63	5.11	6.13	(7)	
	HDD10.0	2211	544	448	312	114	17	0	0	0	4	76	239	457	(8)	
	HDD18.3	4192	802	681	564	339	161	38	9	16	90	292	484	716	(9)	
	CDD10.0	1339	0	0	6	26	129	266	354	316	192	45	4	0	(10)	
	CDD18.3	277	0	0	1	1	14	54	105	73	27	2	0	0	(11)	
	CDH23.3	2688	0	0	4	24	184	521	1016	630	284	25	0	0	(12)	
	CDH26.7	785	0	0	0	5	53	148	339	166	72	3	0	0	(13)	
(14) Wind	WSAvg	2.7	2.9	2.9	2.9	3.3	3.0	2.5	2.3	2.0	2.2	2.8	2.9	2.8	(14)	
(15) Precipitation	PrecAvg	807	32	25	44	70	97	109	104	89	84	68	47	37	(15)	
	PrecMax	990	81	45	86	128	164	211	263	243	171	149	116	80	(16)	
	PrecMin	687	5	8	4	15	41	43	29	22	16	6	4	6	(17)	
	PrecStd	79	19	11	24	27	33	51	56	42	42	38	29	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	10.9	22.4	27.0	31.2	32.5	34.1	32.5	32.0	26.8	18.9	12.2	(19)	
		MCWB	4.3	7.6	15.9	16.5	20.7	22.9	24.3	23.7	23.0	20.0	13.3	10.5	(20)	
	2%	DB	4.0	7.0	16.9	22.4	27.8	30.1	32.2	30.2	28.7	22.7	15.2	7.3	(21)	
		MCWB	2.0	4.2	11.9	14.1	18.8	21.4	23.8	22.3	21.7	17.1	11.5	5.0	(22)	
	5%	DB	2.4	4.0	12.4	18.9	25.1	27.9	30.1	28.0	27.1	20.0	12.5	4.9	(23)	
		MCWB	1.2	2.2	8.3	11.8	16.8	20.7	22.0	21.0	19.9	15.5	8.9	3.0	(24)	
	10%	DB	1.2	2.4	8.7	16.3	22.5	26.4	28.0	27.1	24.2	17.3	9.8	2.7	(25)	
		MCWB	0.4	1.0	5.6	9.6	15.8	19.1	21.0	20.3	18.6	13.5	7.4	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	7.8	16.7	18.1	22.7	25.0	27.0	25.5	24.0	21.7	14.5	11.1	(27)	
		MCDB	7.3	9.5	21.2	23.6	28.2	29.8	31.4	29.7	28.7	24.9	17.4	12.2	(28)	
	2%	WB	2.5	4.5	12.4	15.2	20.1	23.5	25.0	23.9	22.5	18.7	12.1	5.6	(29)	
		MCDB	4.1	6.5	14.9	20.8	25.4	28.1	29.7	28.2	26.9	21.5	14.6	6.7	(30)	
	5%	WB	1.3	2.7	8.7	12.7	18.7	22.2	23.7	22.7	21.2	16.1	9.3	3.1	(31)	
		MCDB	2.2	4.1	12.4	17.7	23.2	26.6	28.2	26.8	25.0	19.3	12.1	4.4	(32)	
	10%	WB	0.3	1.0	5.6	10.5	16.9	20.7	22.5	21.5	19.7	13.7	7.3	1.7	(33)	
		MCDB	1.0	2.4	8.8	15.5	21.1	24.4	26.7	25.3	23.4	16.6	9.1	2.7	(34)	
(35) Mean Daily Temperature Range	MDBR		8.1	9.1	9.8	11.3	12.0	11.6	12.0	11.5	12.6	10.4	8.4	7.5	(35)	
	5% DB	MCDBR	8.6	9.6	14.1	17.1	15.5	13.7	13.6	13.0	14.9	14.7	13.1	9.0	(36)	
		MCWBR	7.4	7.4	9.1	9.8	7.6	6.7	6.2	6.2	7.8	9.0	9.0	7.4	(37)	
	5% WB	MCDBR	7.8	8.9	13.4	15.1	13.4	12.3	11.8	11.3	12.5	12.6	11.7	8.0	(38)	
		MCWBR	6.9	7.0	9.2	9.7	7.3	7.0	6.4	6.3	7.3	9.0	9.0	7.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.256	0.313	0.330	0.362	0.402	0.401	0.388	0.392	0.366	0.321	0.294	0.265	(40)	
	taud		2.400	2.215	2.352	2.335	2.267	2.297	2.342	2.327	2.396	2.522	2.512	2.443	(41)	
	Ebn at Noon		880	863	905	899	869	869	875	859	855	859	828	832	(42)	
	Edh at Noon		72	106	107	119	132	128	121	119	102	79	66	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.64	2.72	3.88	4.56	5.47	5.95	6.11	5.21	4.13	2.60	1.65	1.20	(44)	
	RadStd		0.17	0.25	0.41	0.47	0.43	0.43	0.29	0.34	0.31	0.27	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page